PHOTOCHEMISTRY.

* Photochemical rⁿs proceeds by energy from Radiations. (mostly UV & V.L)

* $E = h\nu$
* $E = \frac{hc}{\lambda}$
* $E = hc\bar{\nu}$

* Cosmic rays
* Gamma rays - ejection of inner cell ēs.
* X-rays - ionising radiatⁿ - valence ē ejected out of ēs molecule
* U.V rays - transition of ē from one level to another.
* visible - electronic transition
* Infrared - heating of biomolecules.
* Microwaves - Sts. identificatⁿ of molecule, cooking - actⁿ on H_2O molecules
* Radiowaves - transmission (communicatⁿ) (highest wavelength)

Spectroscopy ⇒ Interactⁿ of radⁿ with matter.

* In photochemical rⁿ; atleast one of the reactants should be activated.

* Photochemical rⁿ leads to increase in free energy. energy of rⁿ transferred into ΔG. (still it occurs)

* photochemical rⁿ; temperature have no role. but intensity of radⁿ do have.

⇒ LAWS IN PHOTOCHEMICAL RXNS.

⇒

[1] GROWTHUS - DRAPER LAW.

- Only the absorbed rad^n leads to a photochemical rxn.

[2] STARK - EINSTEEN LAW.

- In a photochemical rxn ; one molecule/mole of reactant absorb one photon and give rise to pdt formation.

$$A + h\nu \longrightarrow \text{pdt}$$

$$\text{Quantum yield } (\phi) = \frac{\text{no. of molecules reacted.}}{\text{no. of photons absorbed.}}$$

ideal ph rxn ; $\phi = 1$

$\phi > 1$ more reactants consumed

$\phi < 1$ not efficient.

$\Phi > 1$

1. R^n subsequent to 1° R^n.

$A-B + h\nu \longrightarrow A + B$

$A + A-B \longrightarrow A_2B + B$

at the expense of one photon; two molecules consumed.

2. chain R^n.

$A-A + h\nu \longrightarrow 2A^*$

$A^* + B_2 \longrightarrow AB + B^*$

$B^* + A_2 \longrightarrow AB + A^*$

$\Phi < 1$

1. Reactivation

$A + h\nu \longrightarrow A^*$

$A^* \longrightarrow A + h\nu'$

* excitation and deexcitation occurs.

2. Recombination.

$A-B + h\nu \longrightarrow A + B$

$A + B \longrightarrow A-B$

* either exo or endothermic only.

just like $A \overset{h\nu}{\rightleftharpoons} B$

3 BEER - LAMBERT LAW

UV & V.L



$I \longrightarrow dI$

* when I changes by dI

$$-\frac{dI}{dx} \propto CI$$

$-\frac{dI}{dx}$ → decrease in Intensity with thickness.

CI → Intensity transmitted of rad^n passed; concentration of soln.

$$-\int_{I_0}^{I} \frac{dI}{I\,\text{abs}} = \int_0^{l} C\alpha\, dx$$

I_0 = Intensity of incident rad

I = " of transmitted rad^n.

$$-\ln I\Big)_{I_0}^{I} = C\alpha x\Big)_0^{l}$$

$$2.303 \log\left(\frac{I_0}{I}\right) = C\alpha l$$

$$\varepsilon = \frac{\alpha}{0.303}$$

Absorbance :

$$A = \log\left[\frac{I_0}{I}\right] = \varepsilon C l$$

$l \Rightarrow$ path length.

* Only applicable where

Soln $\Rightarrow$ Very dilute

rad^n $\Rightarrow$ monochromatic.

• if I_0 not given, assume it is 100

Q: A substance when dissolved in water at 10^{-3} M con. absorbed 10% of incident rad^n in a path of 1 cm length. Calculate the con. of the soln which can absorb 90% of the same rad^n.

$C = 10^{-3}$ M

$dx = l = 1$ cm

$I_0 = 100$

$I = 90$

$$\log(I_0/I) = \varepsilon C l$$

$$\log(100/90) = \varepsilon \times 10^{-3} \times 10^{-2}$$

$$0.04 = \varepsilon \times 10^{-5}$$

$$\varepsilon = \frac{0.04}{10^{-5}} = 4575 \text{ M}^{-1}\text{m}^{-1}$$

$C' = ?$ $\quad I_0 = 100 \quad I = 10$

$$\log (100/10) = \varepsilon C' l$$

$$C' = \frac{1}{4000 \times 10^{-2}}$$
4575

$$= 0.0025 \text{ M}$$

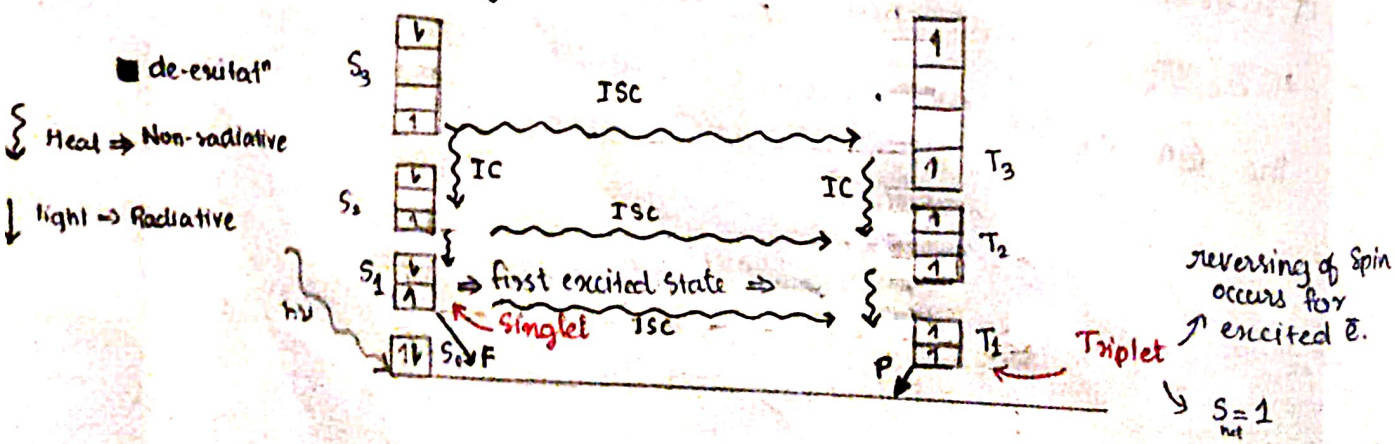
$$= 0.0218 \text{ M}$$

$$A_1 = \log \left(\frac{I_0}{I}\right)_1 = \varepsilon C_1 l$$

$$A_2 = \log \left(\frac{I_0}{I}\right)_2 = \varepsilon C_2 l$$

$$\frac{C_1}{C_2} = \frac{\log \left(I_0/I\right)_1}{\log \left[I_0/I\right]_2}$$

⇒ Jablonski Diagram.



$S_{net} = 1$

At ground state ; when paired .

net spin = 0

Spin multiplicity ; $S = (2s+1)$

at ground state ⇒ $S = (0+1) = 1$ ⇒ Singlet

* ISC ⇒ Intersystem Crossing :- radiationless non-radiative de-excitation b/w states of diff Spin multiplicity.
* IC ⇒ Internal conversion :- non-radiative de-excitation b/w states of same spin multiplicity.

* F ⇒ Flourescence :- radiative de-excitation from first Singlet (S_1) to. S_0.
when light source cut off; immediatly stops.

* P ⇒ phosphorescence (Longer) :- * radiative de-excitation from first Triplet (T_1) to T_0

* also called delayed flaorescence.
:- when source cut off; it retains for some time and then stops.

This happens bcuz: the $\bar{e}$ in same spin $\boxed{\uparrow}\boxed{\uparrow}$ T_1 in triplet Excited State have to first change it's spin and then deexcites. So little time is delayed than that of flaorescense.

⇒ Chemiluminescence.

* The emission of light, at ordinary temp as a result of chemical rxn – chemiluminescence.
* reverse of photochemical rxn.
* As the part of heat of rxn excites some of the molecules and these molecules while reverting to ground state emit radiation usually in visible region. - so called radiations – "cold light"

eg:- ① glow of fire flies - aerial oxidn of luciferon (a protein) in the presence of enzyme lucaferase.

photosensitization.

* The chemical rⁿ occurs by the absorption of light radiatⁿ by a third substance which transfers the absorbed energy to reactants.

* photosensitizer :- a spcs which can both absorb and transfer radiant energy for activation of reactant molecule.

eg: Hg, Cd, Zn, benzophenone, sulphur dioxide.

* Action of chlorophyll (600-700 nm) in permitting CO_2 & H_2O to react in presence of sunlight